



1. Bayesian Inference (40%):

"General Y said that since the interception rate of a Sky Bow missile is 70%, the interception rate of 3 sky bow missiles is 210%." This sentence is obviously incorrect! Let's adopt Bayesian inference to help the General to figure out it.

The problem formulation is as follows (see Fig.1 (a)): Let's consider the action (e.g., Sky Bow missile) and measurement uncertainties (e.g.m Lidar detection). There is a Hidden Markov Model in Fig. 1(b). Assume there are an enemy missile (E missile) and 3 Sky Bow missiles (SB missile). Let's fire 3 SB missiles sequentially at time $i = 1 \sim 3$. Assume the hidden variables (E missile is active) X_i to be boolean ($X_i \in \{1, 0\}$), where $i = 1 \sim 3$. $X_i = 1$ means that E missile is active in time i . Assume the action term u_i to be boolean ($u_i \in \{1, 0\}$). $u_1 = 1$ means that the 1st SB missile is fired. Assume the Lidar measurement variables Z_i to be boolean ($z_i \in \{1, 0\}$), where $i = 1 \sim 3$. $Z_i = 1$ means that E missile is detected in time i . Let $P(X_0 = 1) = P(X_0 = 0) = 0.5$. Let the transition matrix $P(X_{t+1}|X_t, u_t)$ and sensor matrix $P(Z_t|X_t)$ be given by

$$T = \begin{bmatrix} 0.3 & 0.7 \\ 0 & 1.0 \end{bmatrix}, Z = \begin{bmatrix} 0.8 & 0.2 \\ 0.1 & 0.9 \end{bmatrix}$$

, where in the T matrix,

$$T_{11} = P(X_{t+1} = 1|X_t = 1, u_t = 1); T_{12} = P(X_{t+1} = 0|X_t = 1, u_t = 1);$$

$$T_{21} = P(X_{t+1} = 1|X_t = 0, u_t = 1); T_{22} = P(X_{t+1} = 0|X_t = 0, u_t = 1);$$

and in the Z matrix,

$$Z_{11} = P(Z_t = 1|X_t = 1); Z_{12} = P(Z_t = 0|X_t = 1);$$

$$Z_{21} = P(Z_t = 1|X_t = 0); Z_{22} = P(Z_t = 0|X_t = 0);$$

Consider the sequences of measurements $Z_{1:3} = \{1, 1, 1\}$

Please answer the following questions.

1) Derive the matrix form of smoothed estimation via aforementioned notations. (20%)

Let the state be ordered as $[x_t=1, x_t=0]^T$. Define the forward, backward, and smoothed vectors as $\alpha_t = P(x_t | z_{1:t}) \in \mathbb{R}^2$, $\beta_t = P(z_{t+1:T} | x_t) \in \mathbb{R}^2$, $\gamma_t = P(x_t | z_{1:T}) \in \mathbb{R}^2$.

For each observation $z_t \in \{1, 0\}$, define the likelihood vector $b_t = \begin{bmatrix} P(Z_t=1|X_t=1), P(Z_t=1|X_t=0) \\ P(Z_t=0|X_t=1), P(Z_t=0|X_t=0) \end{bmatrix}^T$
 $= \begin{cases} [0.8, 0.1]^T, & z_t=1 \\ [0.2, 0.9]^T, & z_t=0. \end{cases}$

and let $D(b_t)$ denote the diagonal matrix with the components of b_t on the diagonal: $D(b_t) = \begin{bmatrix} b_t(1) & 0 \\ 0 & b_t(2) \end{bmatrix}$.

The forward procedure consists of prediction, update, and normalization steps: $\begin{cases} \text{Prediction: } \tilde{\alpha}_t = T^T \alpha_{t-1}, \\ \text{Update: } \bar{\alpha}_t = D(b_t) \tilde{\alpha}_t \\ \quad \quad \quad = D(b_t) T^T \alpha_{t-1}, \\ \text{Normalization: } \alpha_t = \frac{\bar{\alpha}_t}{\mathbf{1}^T \bar{\alpha}_t} \\ \quad \quad \quad = \frac{D(b_t) T^T \alpha_{t-1}}{\mathbf{1}^T D(b_t) T^T \alpha_{t-1}} \end{cases}$

where $\mathbf{1} = [1, 1]^T$ and the initial state is $\alpha_0 = [0.5, 0.5]^T$. The normalization constant $c_t = \mathbf{1}^T D(b_t) T^T \alpha_{t-1}$ is recorded for the backward pass.

The backward message starts from $\beta_T = \mathbf{1}$ and propagates recursively as $\beta_t = \frac{T D(b_{t+1}) \beta_{t+1}}{c_{t+1}}$, $t = T-1, T-2, \dots, 0$, where c_{t+1} is the normalization constant from the corresponding forward step. This scaling maintains numerical stability and ensures $\mathbf{1}^T (\alpha_t \odot \beta_t) = 1$.

The smoothed marginal distribution is obtained by the elementwise product of forward and backward messages:

$$\gamma_t = \alpha_t \odot \beta_t, \text{ or equivalently, without pre-scaling, } \gamma_t = \frac{\alpha_t \odot \beta_t}{\mathbf{1}^T(\alpha_t \odot \beta_t)}.$$

Combining all equations, the complete forward-backward smoothing algorithm in matrix form is

$$\begin{cases} \alpha_0 = [0.5, 0.5]^T \\ \alpha_t = \frac{D(b_t)^T \alpha_{t-1}}{\mathbf{1}^T D(b_t)^T \alpha_{t-1}}, \quad t=1,2,3, \\ \beta_3 = \mathbf{1}, \\ \beta_t = \frac{T D(b_{t+1}) \beta_{t+1}}{c_{t+1}}, \quad t=2,1,0, \\ \gamma_t = \alpha_t \odot \beta_t, \quad t=1,2,3. \end{cases}$$

Because all three observations are $z_t = 1$, $D(b_t) = \begin{bmatrix} 0.8 & 0 \\ 0 & 0.1 \end{bmatrix}$, $T = \begin{bmatrix} 0.3 & 0.7 \\ 0 & 1.0 \end{bmatrix}$, $a_0 = \begin{bmatrix} 0.5 \\ 0.5 \end{bmatrix}$, $\beta_3 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.

Substituting these matrices into the equations above yields the exact numeric recursion for the fitting and smoothing probabilities of the enemy missile being active at each time step.

2 - 5) Compare and explain each algorithm's performance. (4%)

All four models (NB, Perceptron, Logistic, AdaBoost) achieve 100% Train/Test. The data are nearly linearly separable in $\{x, y, z\}$; Perceptron converges in 1-2 epochs, Logistic's cross-entropy $\rightarrow 0$, NB works despite independence assumptions, and Adaboost enlarges the margin.

Pick Logistic (probabilities) or AdaBoost (robust) as the final model.

3 - 5) Compare and explain each algorithm's performance. (8%)

Naïve Bayes: $\text{Acc} \approx 0.74 \sim 0.75$, $\text{Recall} \approx 0.99$, $\text{Precision} \approx 0.26 \rightarrow$ many false positives (independence assumption violated by correlated geometric features).

Perceptron: $\text{Acc} \approx 0.89 \sim 0.90$, $\text{Prec} \approx 0.47$, $\text{Rec} = 1.00$; mistake rate plateaus $\rightarrow$ boundary not fully linear, tends to over-predict "leg".

Logistic: $\text{Acc} \approx 0.91$, but predicts all negatives with threshold 0.5 (class imbalance/overlap). With threshold tuning it would improve.

AdaBoost: Best! - Train ≈ 0.974 , Test ≈ 0.988 , $\text{F1} \approx 0.935$; stumps form a piecewise non-linear boundary well-suited to radius/width/circularity relations.

Conclusion: Use AdaBoost for laser; Logistic becomes competitive after threshold turning.

